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Figure 1 Expression of Sox9 in the presence and absence of miRNA-145 or anti-miRNA-145 following induction of chondrocyte differentiation

Osteoporosis, a disease marked by decreased bone density and mass, is similar to OA in that it is a degenerative bone disorder correlated with old age. Which two skeletal system functions would be most significantly impaired by osteoporosis and OA, respectively?

- A. Structural support and mobility [76%]
- B. Mobility and phosphate release [1%]
- C. Structural support and calcium storage [17%]
- D. Calcium storage and phosphate release [3%]

Correct
77% answered correctly

Explanation:

Skeletal system functions	
Mobility	• Provides a framework for muscle attachment to generate body movement
Structural support & protection	• Provides weight-bearing support • Protects internal organs from injury
Blood cell production (hematopoiesis)	• Red bone marrow contains hematopoietic stem cells that give rise to red & white blood cells
Mineral reservoir	• Bone stores phosphate & calcium, which are deposited & released based on physiological requirements

The primary functions of the skeletal system include **mobility**, **structural support**, **physical protection** of internal organs (eg, the skull protects the brain, the ribcage protects the lungs), **mineral storage** (primarily phosphate and calcium), **blood cell production** (hematopoiesis) in red bone marrow, and energy storage (fat stored in the yellow marrow).

In osteoporosis, *loss of bone density and mass* leads to weakened bones that are more brittle and susceptible to fracture. The passage states that the formation of bony growths (osteophytes) and narrowing of the joint space in OA *limits* *movement*. Therefore, based on the given scenario, **osteoporosis would** significantly **compromise** the structural integrity of bone and its ability to provide **support** (eg, fracture of weight-bearing bones that function as support pillars), and OA would significantly **limit mobility**.

(Choices B, C, and D) Calcium and phosphate are generally stored as calcium phosphate crystals in the inorganic portion of bone. Per the passage, OA is marked by the erosion of articular cartilage, damage to the underlying cortical bone, and the formation of osteophytes (small, bony growths). Because osteophytes are *small* and likely contain very low levels of phosphate and calcium relative to bones of the skeletal system, their formation is *unlikely* to significantly impair phosphate release from bone. In addition, there is *no* information in the passage to suggest loss of bone density and mass associated with OA, indicating that the calcium-storing ability of bone is *not likely impaired* in OA. In contrast, calcium storage *would* likely be impaired in cases of osteoporosis due to decreased bone mineralization.

Educational objective:
The skeletal system facilitates mobility, provides the body with structural support, is the site of hematopoiesis, protects the body's internal organs, and serves to store fat, calcium, and phosphate.

Time spent: 0 sec	**QID**:401061
Subject: Biology	**System:** Musculoskeletal System

Osteoarthritis (OA) is a disorder marked by the deterioration of articular (hyaline) cartilage, the connective tissue lining the ends of bones at most movable joints. The progressive erosion of articular cartilage in OA is thought to be caused by an inflammatory response induced by years of biomechanical stress due to excessive compression and friction. This response involves the release of proteolytic enzymes that break down cartilaginous proteins. The progressive breakdown of articular cartilage ultimately results in complete exposure of the cortical portion of the underlying subchondral bone with narrowing of the joint space. Repeated and abnormal joint stress can also cause the development of osteophytes (or small, bony growths) that form on the subchondral bone. These degenerative processes cause pain and restrict joint movement.

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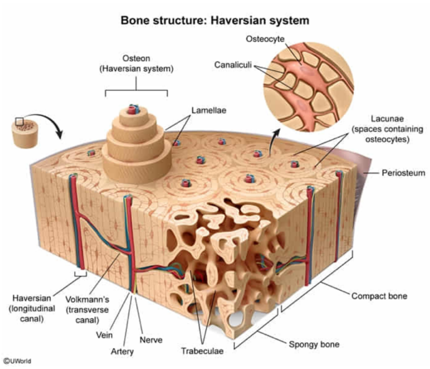
Microscopic examination of tissue samples obtained from the type of bone that becomes exposed in advanced OA is most likely to reveal all the following EXCEPT:

- A. Volkmann canals connecting haversian canals. [13%]
- ✔ B. mature chondrocytes. [65%]
- C. canaliculi. [8%]
- D. osteocytes situated within lacunae. [12%]

Correct
65% answered correctly

Explanation:



Based on the passage, the degradation of articular cartilage in advanced OA can ultimately result in complete exposure of the cortical portion of the underlying subchondral bone. A shell of cortical (compact) bone generally surrounds cancellous (spongy) bone. Therefore, the **compact bone becomes exposed** in advanced OA.

Compact (hard) bone is organized into structural units called **osteons**, or haversian systems, which are made up of

lamellae (concentric rings of bone matrix) that surround a central **haversian canal**, a cylindrical channel that runs parallel to the long axis of bone and through which blood vessels and nerves traverse. **Volkmann canals**, which run perpendicular to the long axis of bone, allow the passage of blood vessels and nerves between different haversian canals **(Choice A)**.

Other cellular components of bone include *osteogenic (osteoprogenitor) cells*, *osteoblasts*, *osteocytes*, and *osteoclasts*. During bone remodeling, old bone is resorbed (broken down) by osteoclasts and new bone is deposited by osteoblasts. Osteogenic cells are the mitotically active stem cells in bone that initially differentiate into osteoblasts. The osteoblasts then secrete the proteins that form the unmineralized bone matrix called **osteoid**. Although it is primarily composed of collagen, osteoid eventually becomes mineralized through the precipitation of calcium salts on the surfaces of its collagen fibers. These deposited calcium salts mature into hydroxyapatite crystals, the mineral responsible for bone hardness.

Osteoblasts continue to build successive concentric layers of bone but as the osteoid mineralizes, some osteoblasts become trapped within **lacunae** (spaces) in the lamellar matrix and are known as (mitotically inactive) osteocytes at this stage. *Within each osteon* of compact bone, lacunae connect to one another via microscopic channels called **canaliculi**, which allow osteocyte waste exchange and nutrient delivery **(Choices C and D)**.

Chondrocytes, however, make up the cellular component of cartilage, *not* bone.

Educational objective:
Compact bone is organized into concentric rings of bone matrix called lamellae. The entire unit of concentrically arranged lamellae surrounding a central haversian canal is known as an osteon, or a haversian system. Within each osteon, lacunae (spaces containing osteocytes) connect to one another via microscopic channels called canaliculi, which allow osteocyte waste exchange and nutrient delivery.

Time spent: 0 sec	**QID**:401062
Subject: Biology	**System:** Musculoskeletal System

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Figure 1 Expression of Sox9 in the presence and absence of miRNA-145 or anti-miRNA-145 following induction of chondrocyte differentiation

Which of the following statements is NOT true about hyaline cartilage?

- A. It lacks innervation. [9%]
- B. It gives rise to bone through endochondral ossification. [25%]
- C. It serves as an attachment that holds bones to muscle. [58%]
- D. It is avascular and receives nutrients from surrounding fluids. [6%]

Correct
58% answered correctly

Explanation:

Cartilage type	Functions
Hyaline	• Reduces friction between bony surfaces to facilitate joint movement • Allows linear bone growth at the epiphyseal plate in childhood • Reinforces respiratory passageways • Supports the external nose
Elastic	• Highly flexible; withstands distortion without damage and reverts to its original shape (eg, external ear)

Fibrous	• Limits movement • Resists compression forces and stretch • Prevents direct contact between bones (eg, between vertebrae of the spine)

Cartilage is a connective tissue made up of cells that secrete a specialized extracellular matrix called chondrin, which contains collagen fibers, proteoglycans, and water. Cartilage may be classified as hyaline, elastic, or fibrous cartilage depending on its composition. The firm but flexible structure of cartilage is resistant to compression and stretching. Hence, most of the cartilage in the body is found in parts that require cushioning (eg, **joint surfaces**, spine) and on the articulating surfaces of bone ends. Specifically, hyaline cartilage is the most abundant cartilage type in the body and is found on the ends of long bones as well as in the ribs, nose, trachea, and larynx. Because cartilage **lacks nerves** (ie, is not innervated) and is **avascular** (ie, lacks its own blood supply), it must receive nutrients and oxygen via diffusion from surrounding fluids or vascularized areas **(Choices A and D)**.

Cartilage also plays a role in certain mechanisms of bone development. The process of **endochondral ossification** uses **hyaline cartilage** as a template for bone deposition **(Choice B)**. During fetal development, hyaline cartilage composing the fetal skeleton is replaced by bone (ie, calcification). In children and adolescents, the epiphyseal (growth) plate of long bones is formed from hyaline cartilage and serves as the site of bone lengthening. In contrast, bone formation can also occur via intramembranous ossification, a process in which bone is formed directly from fibrous connective tissue (*not* hyaline cartilage). Mesenchymal (connective tissue) stem cells within this connective tissue differentiate into osteoblasts and secrete the bone matrix.

Ligaments and tendons are important connective tissue structures at joints. Tendons (*not hyaline cartilage*) tautly anchor muscle to bone. Tendons are strong, fibrous bands of connective tissue that transmit a force generated by contracting muscle to the bone, permitting locomotion (movement). In contrast, ligaments are strong bundles of connective fibers that connect bones to other bones and stabilize and hold structures together.

Educational objective:
Cartilage is a firm but flexible connective tissue that lacks blood vessels and nerves. Chondrocytes (cartilage cells) secrete chondrin, which is the specialized extracellular matrix that makes up cartilage. The most common type of cartilage is hyaline cartilage, which lines the ends of articulating bones and plays a role in bone development.

Time spent:0 sec
Subject:
Biology

QID:401063
System:
Musculoskeletal System

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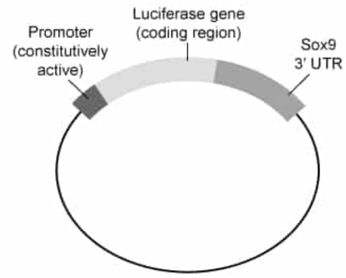
Luciferase is an enzyme responsible for bioluminescence (light emission). Researchers engineered a reporter vector (shown below) that results in the expression of a hybrid transcript containing luciferase and the *Sox9* 3′-UTR.

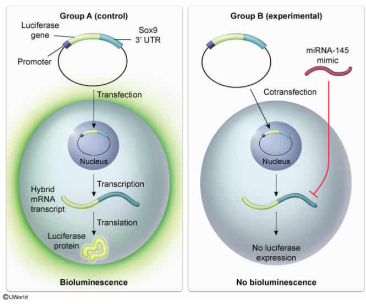


The vector was transfected into two groups of mouse stem cells, Group A (control) and Group B (experimental). Group B cells were also transfected with a miR-145 mimic that was designed to imitate mature endogenous miRNA in the cell. Given this, luciferase bioluminescence will most likely be detected in:

- A. Group A cells only. [62%]
- B. Group B cells only. [17%]
- C. both Group A and Group B cells. [17%]
- D. neither Group A nor Group B cells. [2%]

Correct
62% answered correctly

Explanation:



Reporter genes code for easily identified (ie, fluorescent, luminescent) protein products and are consequently used to **determine** the **effects** of certain **regulatory sequences** or **measure** the **expression of other genes**. For example, reporter genes can be used to determine whether transfection, the incorporation of foreign genetic material into animal cells, has been successful. This is done by including the reporter gene sequence in the vector (the DNA molecule engineered to carry the foreign DNA into the host cell). In addition, in gene expression assays, the reporter gene sequence is fused to the gene or gene region of interest and results in a hybrid transcript, allowing researchers to identify gene expression changes under varying experimental conditions.

The passage states that miRNA-145 regulates Sox9 gene expression by binding the 3′-UTR region of the Sox9 mRNA transcript. Binding of miRNA to an mRNA transcript generally prevents translation of the *entire* transcript. The western blot data in the passage shows that Sox9 expression is absent in the presence of transfected miRNA-145, indicating that miRNA-145 *inhibits* Sox9 expression and consequently prevents chondrogenic differentiation in the cell.

Per the question, the luciferase enzyme is responsible for bioluminescence (emission of light by a living organism). The construct (vector) was engineered to result in a hybrid transcript containing luciferase and the Sox9 3′-UTR. However, treatment with the miRNA-145 mimic would likely *prevent* luciferase expression by binding the Sox9 3′-UTR region and blocking translation of the entire transcript, similar to what occurred when miRNA-145 was introduced in the passage experiment. Therefore, bioluminescence *will likely* be observed in the control group (Group A), which was not treated with the miRNA-145 mimic **(Choice D)**. However, bioluminescence *will likely not* be observed in the experimental group (Group B), which contains the miRNA-145 mimic that blocks translation of the hybrid transcript **(Choices B and C)**.

Educational objective:
A reporter gene allows researchers to study the regulation or expression of other genes. The luciferase gene is a commonly used reporter gene whose protein product catalyzes a reaction that results in bioluminescence, which allows easy expression quantification of the target gene.

Time spent:0 sec
Subject:
Biology

QID:401064
System:
Musculoskeletal System